Performance Report for results

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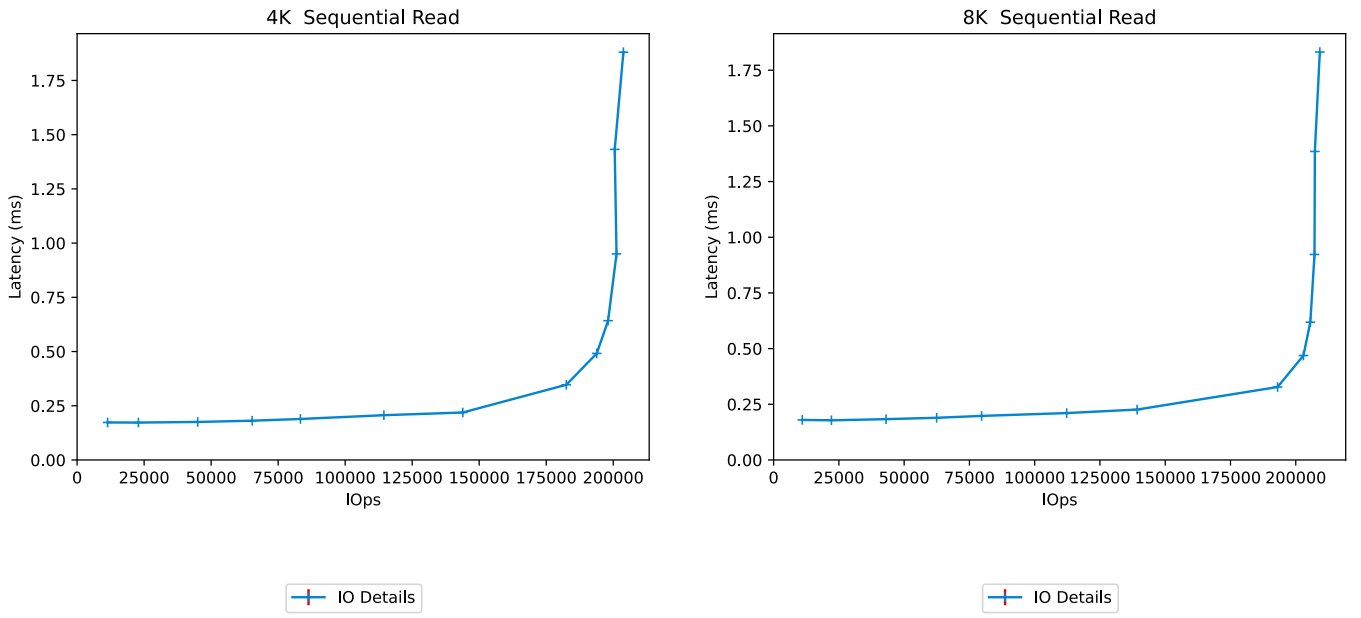
Summary of results for results

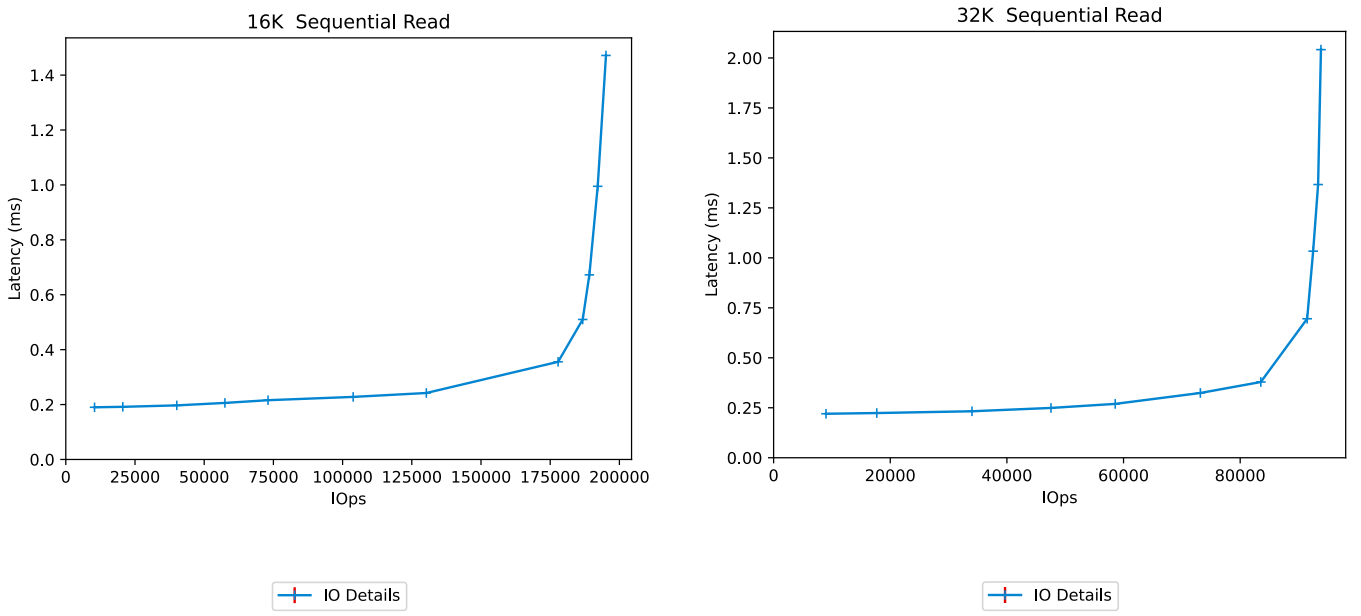
Workload Name	Number of Jobs	Maximum Throughput	Latency (ms)
4K Sequential Read	1	203789 IOps	1.9
8K Sequential Read	1	209214 IOps	1.8
16K Sequential Read	1	195152 IOps	1.5
32K Sequential Read	1	93849 IOps	2.0
64K Sequential Read	1	5308 MB/s	3.2
256K Sequential Read	1	10539 MB/s	1.6
512K Sequential Read	1	11956 MB/s	1.4
1024K Sequential Read	1	11166 MB/s	1.9
2048K Sequential Read	1	10505 MB/s	4.0
4096K Sequential Read	1	13683 MB/s	6.1
4K Sequential Write	1	288390 IOps	5.3
8K Sequential Write	1	219651 IOps	7.0
16K Sequential Write	1	127765 IOps	8.0
32K Sequential Write	1	78227 IOps	9.8
64K Sequential Write	1	2384 MB/s	14.1
256K Sequential Write	1	3115 MB/s	10.7
512K Sequential Write	1	3282 MB/s	10.1
1024K Sequential Write	1	3455 MB/s	14.4
2048K Sequential Write	1	3245 MB/s	40.8
4096K Sequential Write	1	3157 MB/s	41.7
4K Random Read	1	197726 IOps	3.9
8K Random Read	1	189685 IOps	4.0
16K Random Read	1	172756 IOps	3.4
32K Random Read	1	82993 IOps	1.5
64K Random Read	1	5003 MB/s	1.7
256K Random Read	1	11647 MB/s	2.9
512K Random Read	1	14220 MB/s	2.3
1024K Random Read	1	14412 MB/s	2.3
2048K Random Read	1	15558 MB/s	4.3
4096K Random Read	1	15210 MB/s	8.8
4K Random Write	1	14054 IOps	27.3
8K Random Write	1	13239 IOps	38.7
16K Random Write	1	12852 IOps	29.9
32K Random Write	1	10324 IOps	37.2
64K Random Write	1	692 MB/s	36.4
256K Random Write	1	2081 MB/s	32.2
512K Random Write	1	2752 MB/s	12.1
1024K Random Write	1	3162 MB/s	7.8
2048K Random Write	1	3109 MB/s	15.8
4096K Random Write	1	2964 MB/s	44.4
4K 70/30 Random Read/Write	1	35779 IOps	7.2
16K 70/30 Random Read/Write	1	32783 IOps	7.8
64K 30/70 Random Read/Write	1	901 MB/s	18.6
64K 70/30 Random Read/Write	1	1509 MB/s	11.1

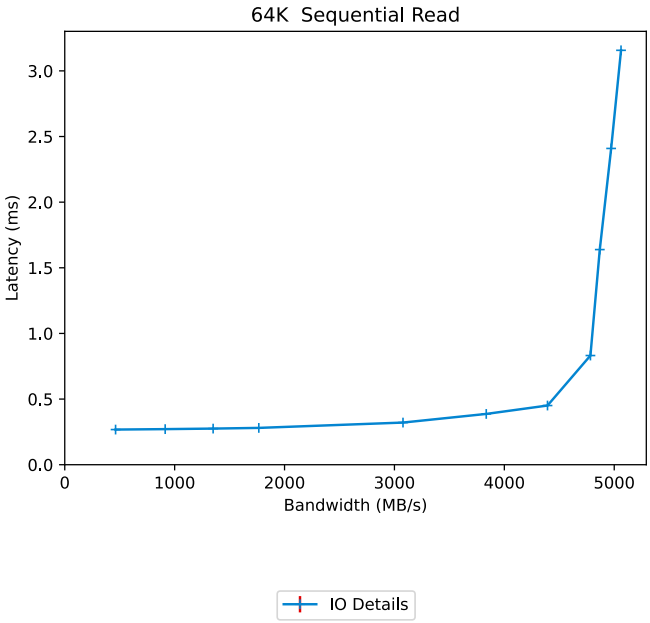
Response Curves

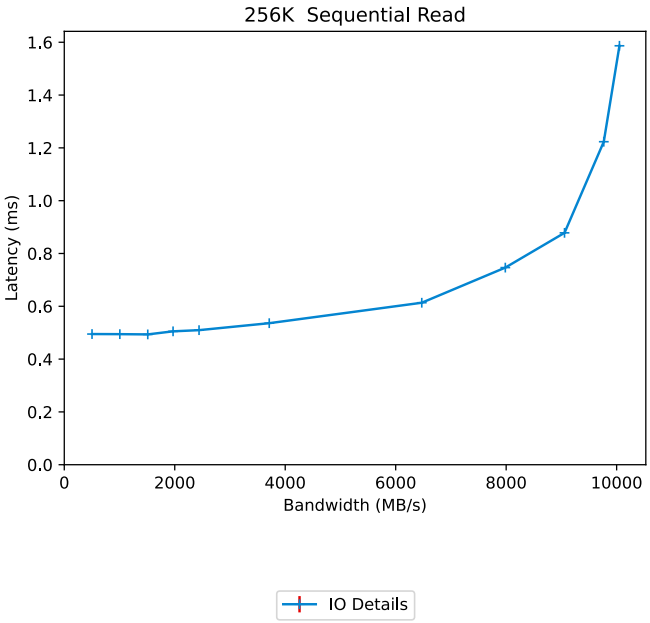
Number of Jobs 1

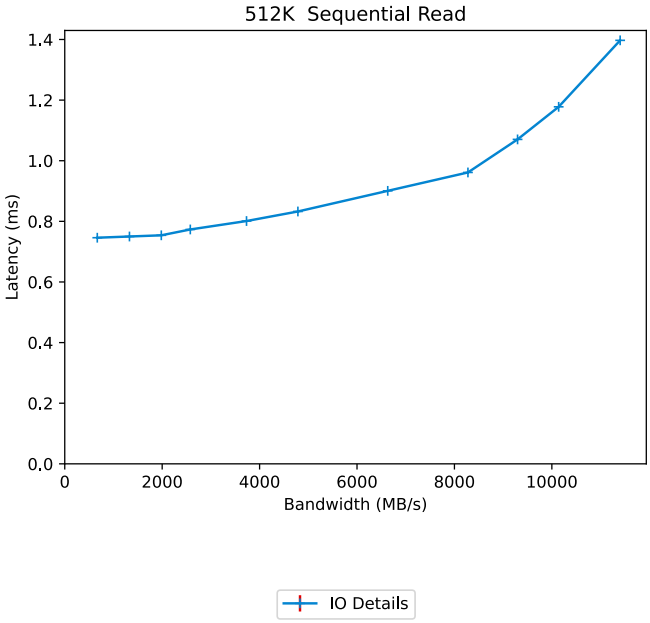
Sequential Read

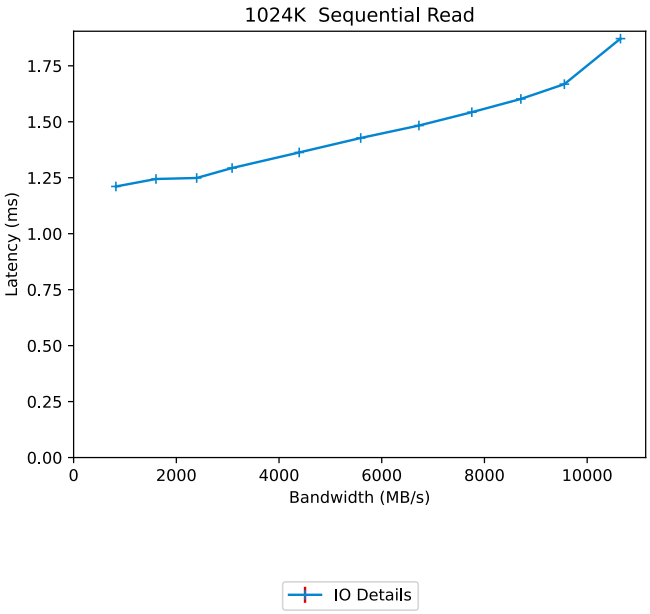


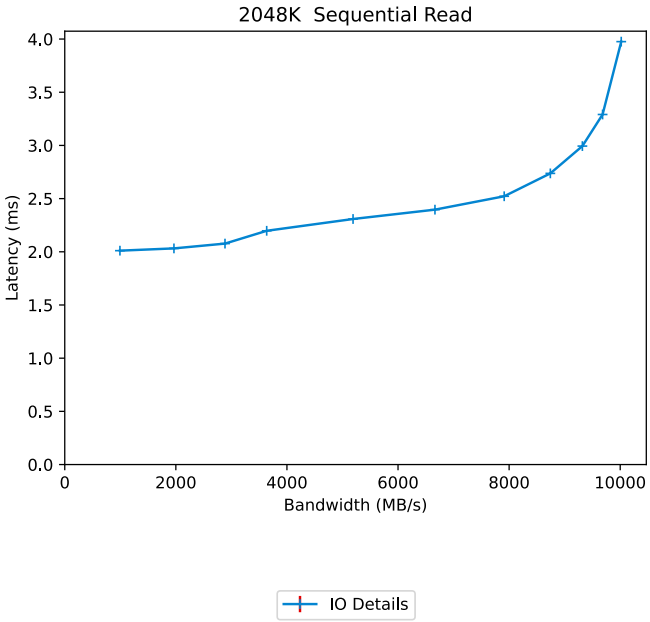


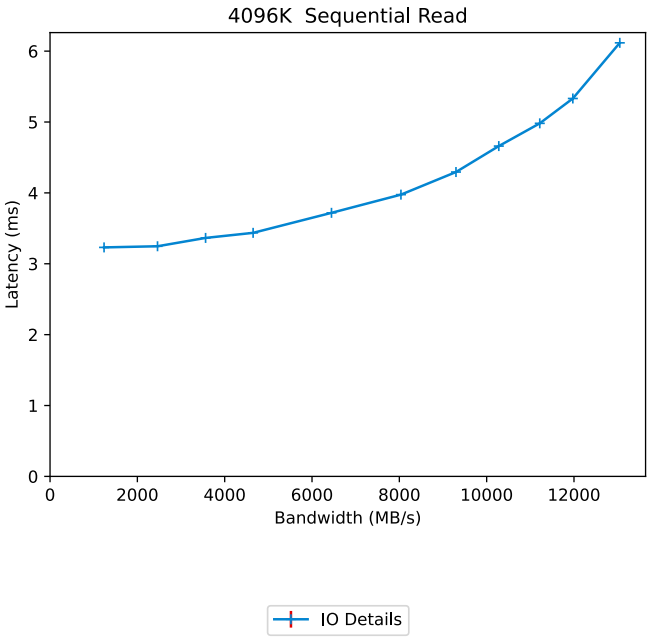

64K Sequential Read
Latency (ms)
Bandwidth (MB/s)
IO Details


256K Sequential Read
Latency (ms)
Bandwidth (MB/s)
IO Details

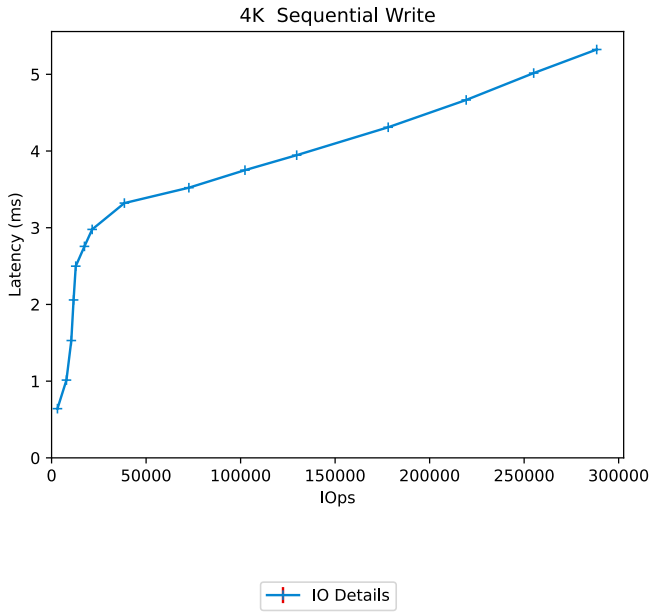

512K Sequential Read
Latency (ms)
Bandwidth (MB/s)
IO Details

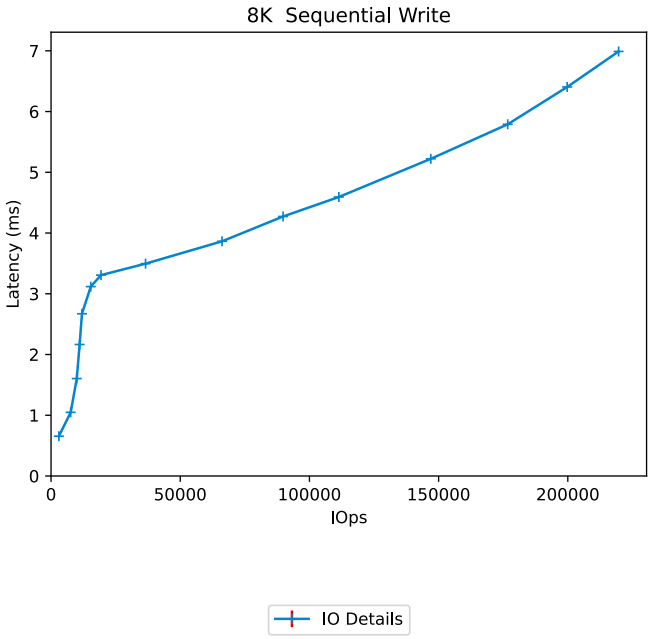

1024K Sequential Read
Latency (ms)
Bandwidth (MB/s)
IO Details

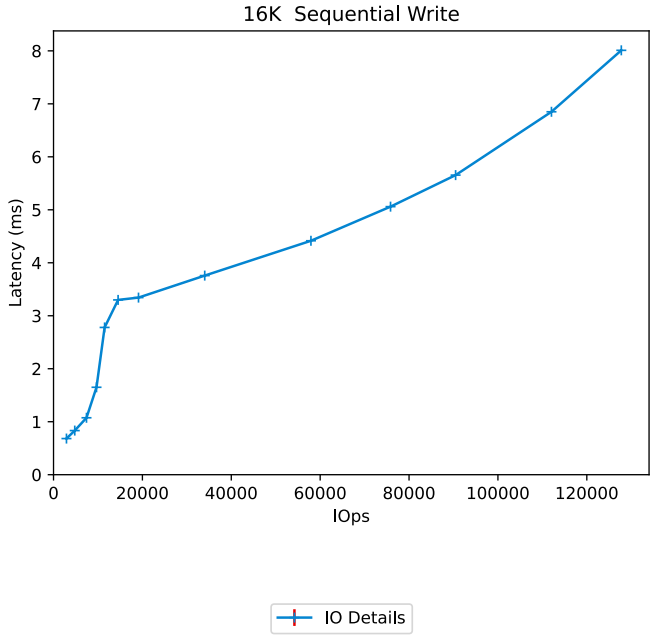

2048K Sequential Read
Latency (ms)
Bandwidth (MB/s)
IO Details

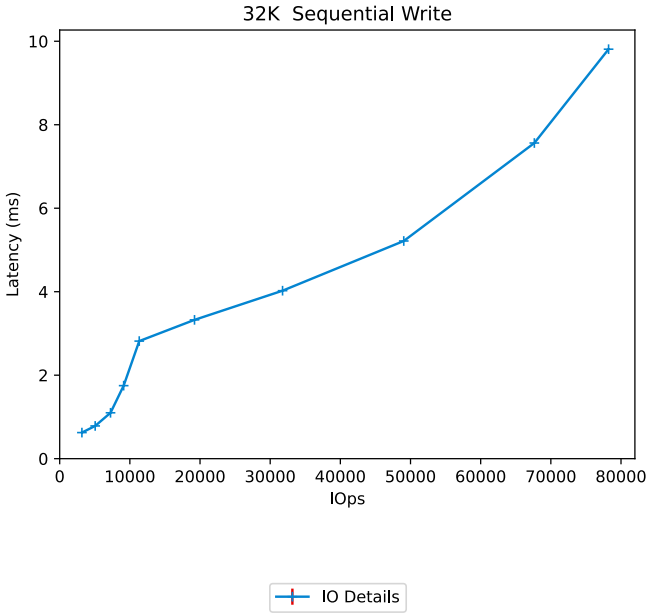

4096K Sequential Read
Latency (ms)
Bandwidth (MB/s)
IO Details

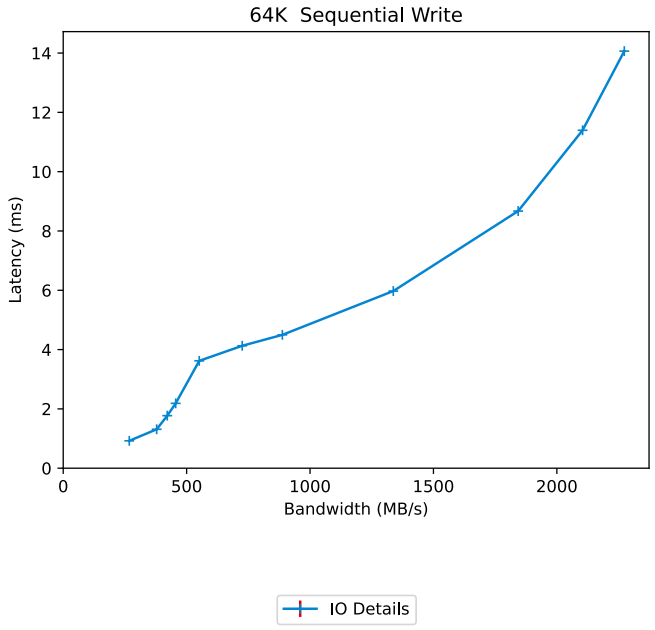
Sequential Write

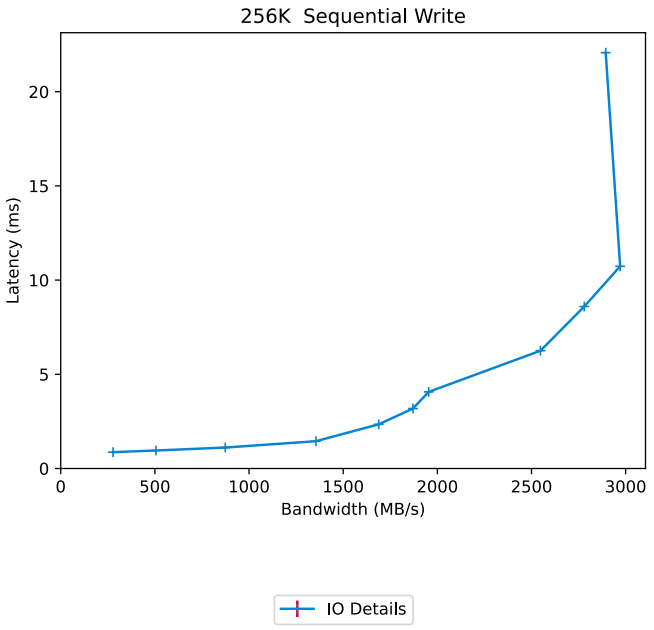


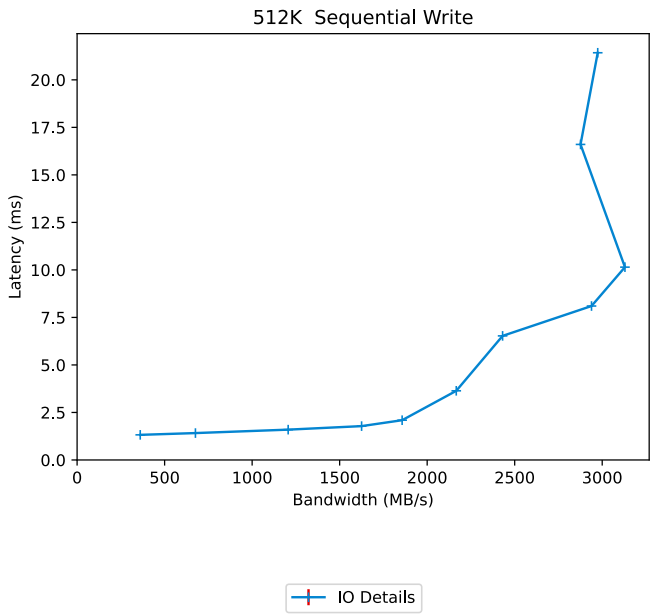


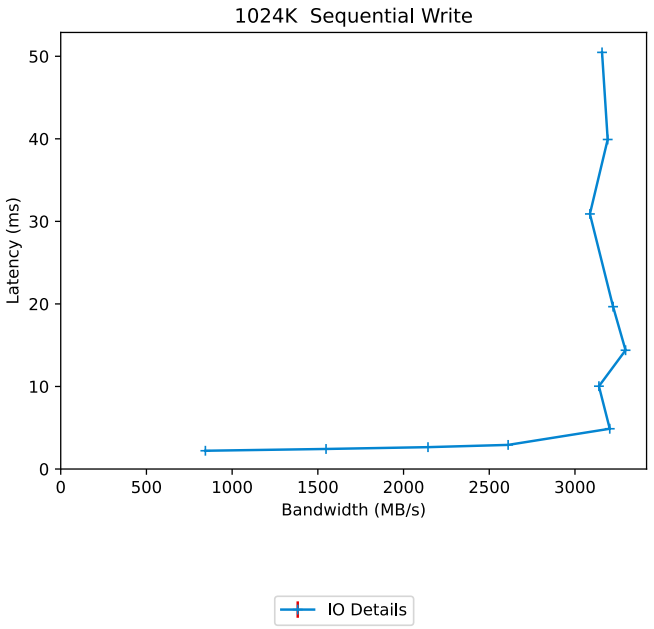


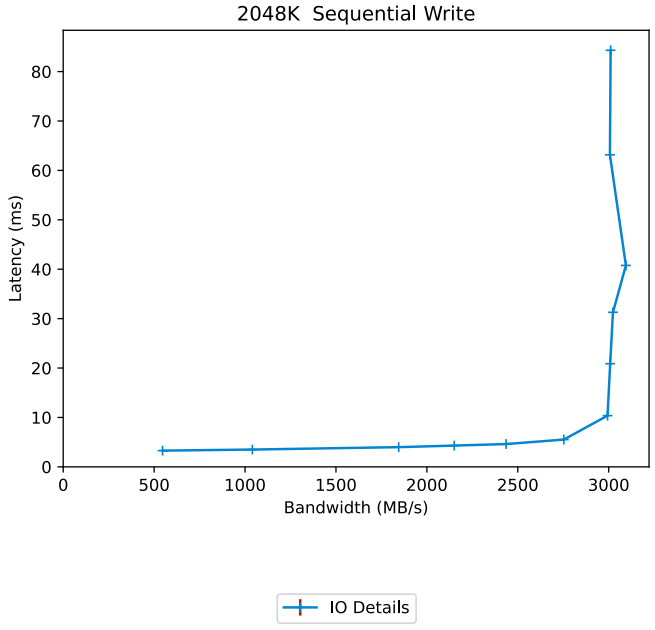


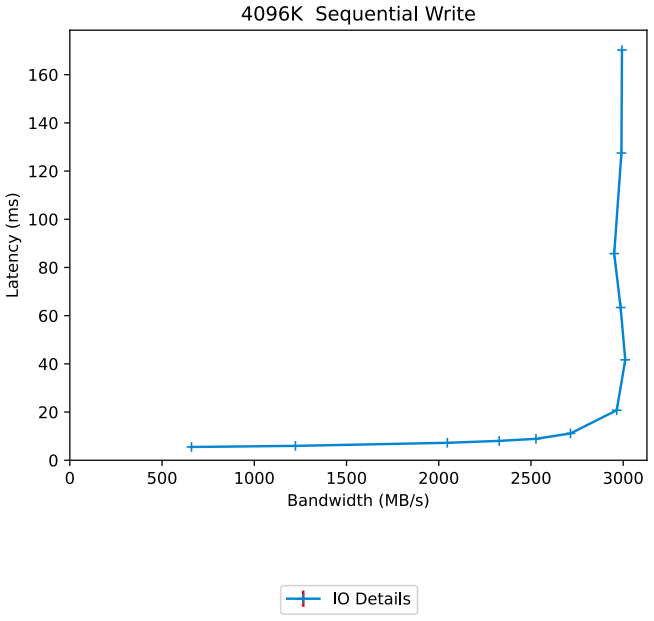

64K Sequential Write
Latency (ms)
Bandwidth (MB/s)
IO Details


256K Sequential Write
Latency (ms)
Bandwidth (MB/s)
IO Details

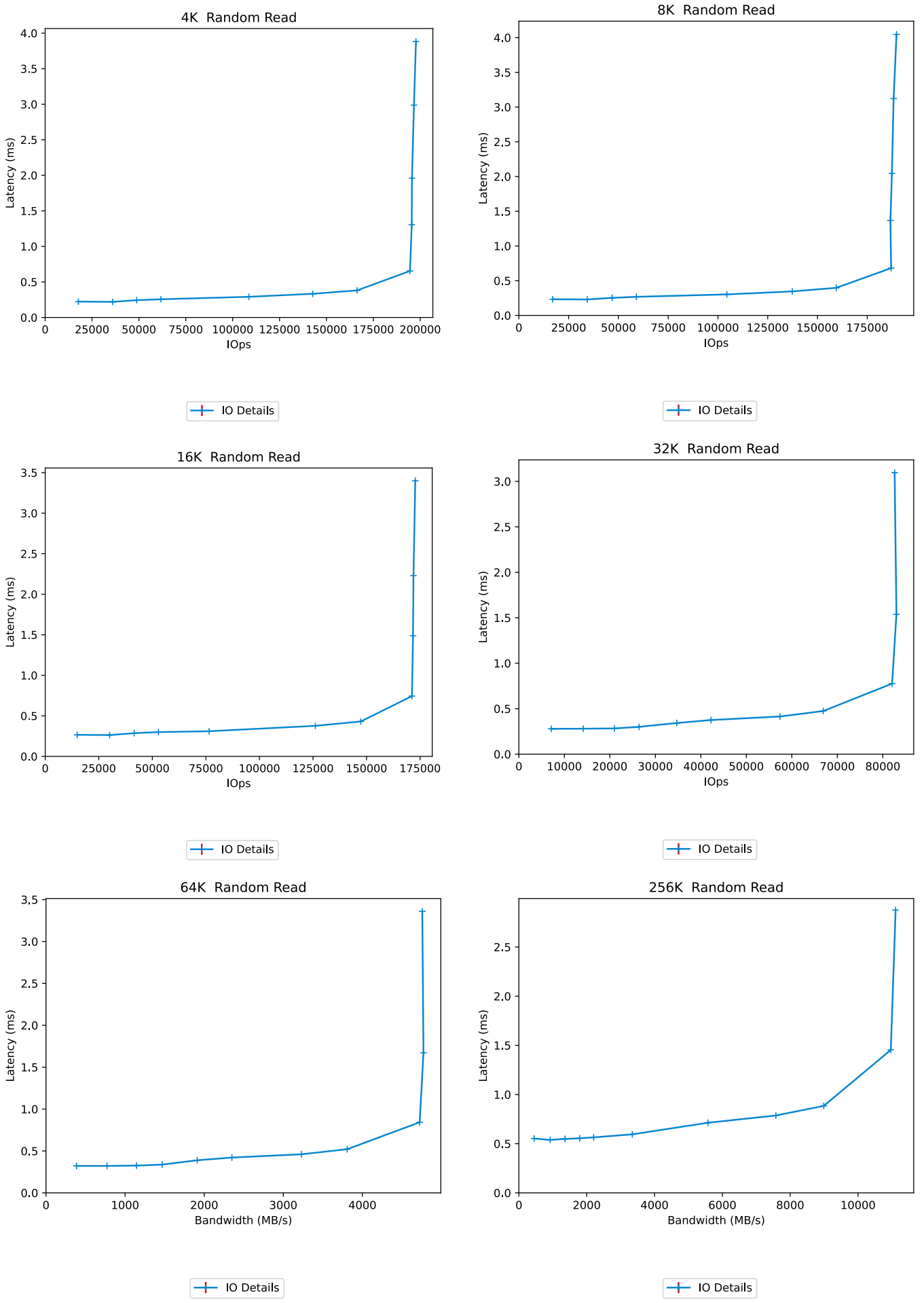

512K Sequential Write
Latency (ms)
Bandwidth (MB/s)
IO Details

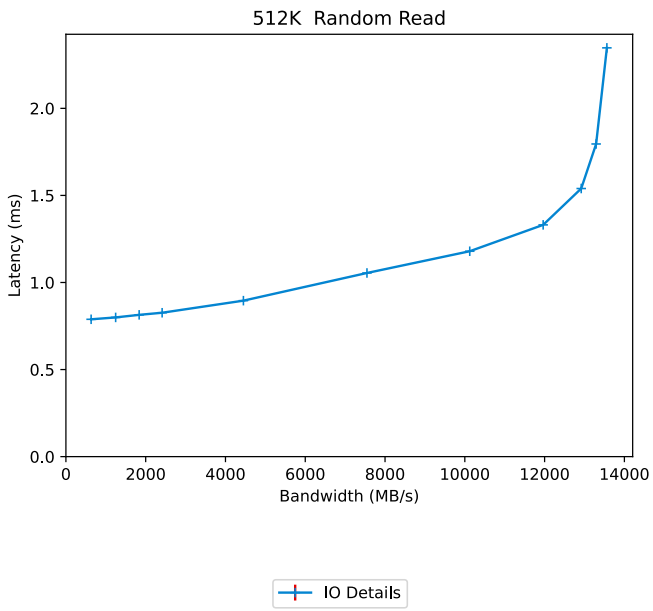

1024K Sequential Write
Latency (ms)
Bandwidth (MB/s)
IO Details

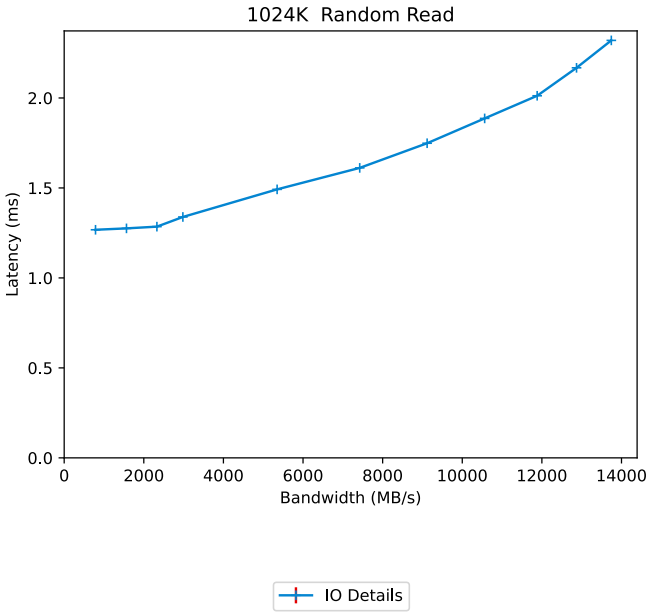

2048K Sequential Write
Latency (ms)
Bandwidth (MB/s)
IO Details

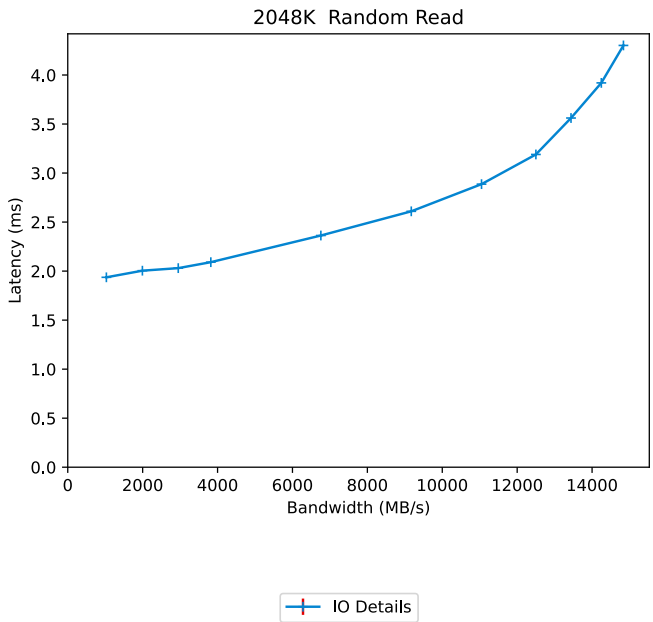

4096K Sequential Write
Latency (ms)
Bandwidth (MB/s)
IO Details

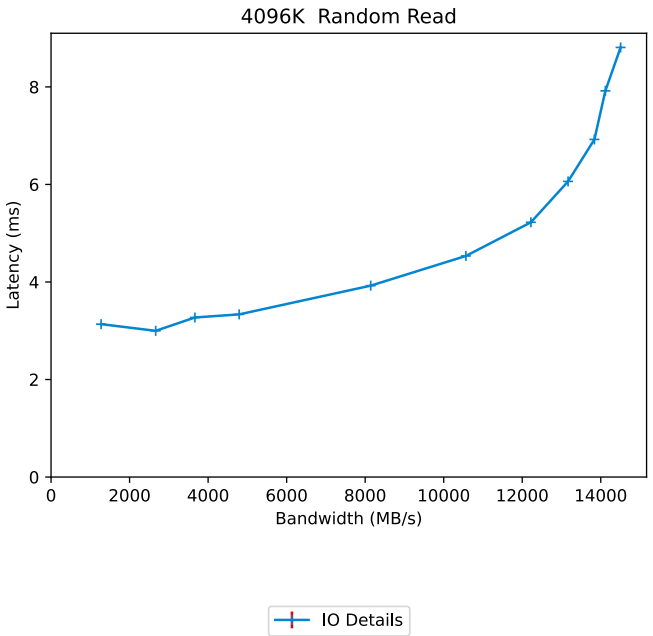
Random Read


4K Random Read
Latency (ms)
IOps
IO Details
8K Random Read
Latency (ms)
IOps
IO Details
16K Random Read
Latency (ms)
IOps
IO Details
32K Random Read
Latency (ms)
IOps
IO Details
64K Random Read
Latency (ms)
Bandwidth (MB/s)
IO Details
256K Random Read
Latency (ms)
Bandwidth (MB/s)
IO Details

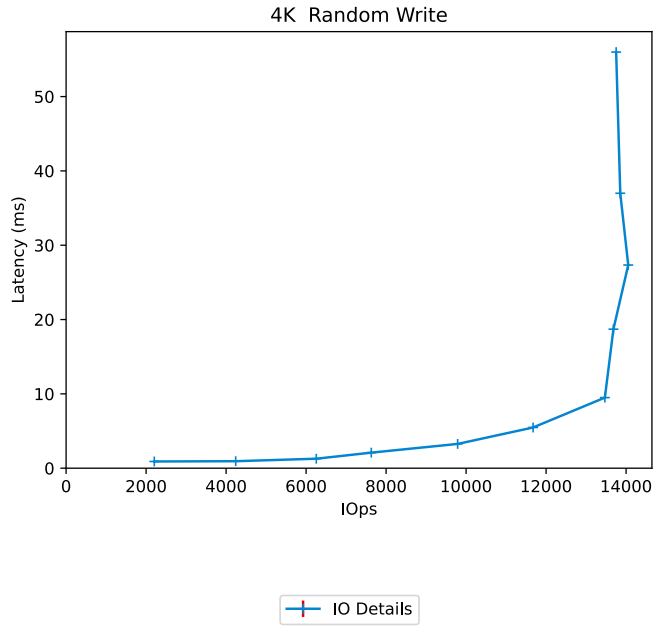


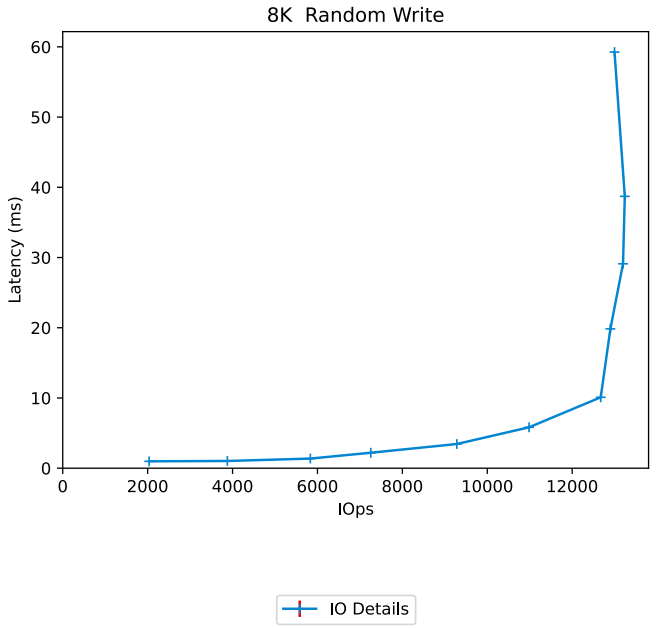


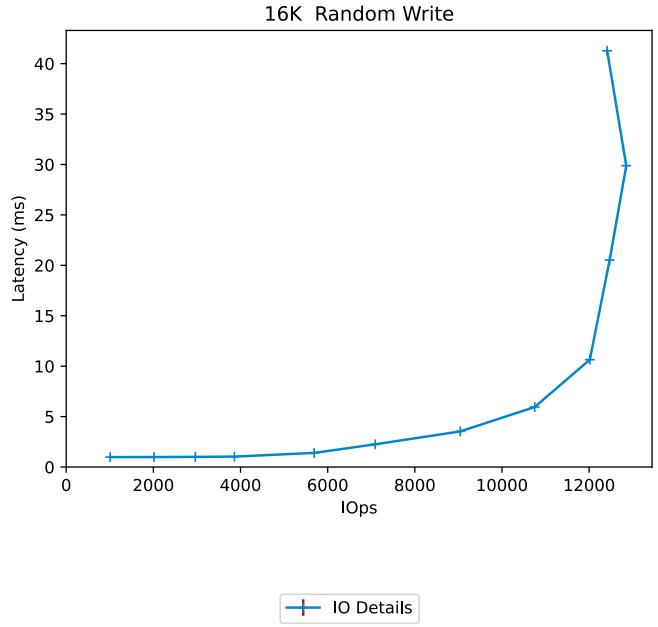


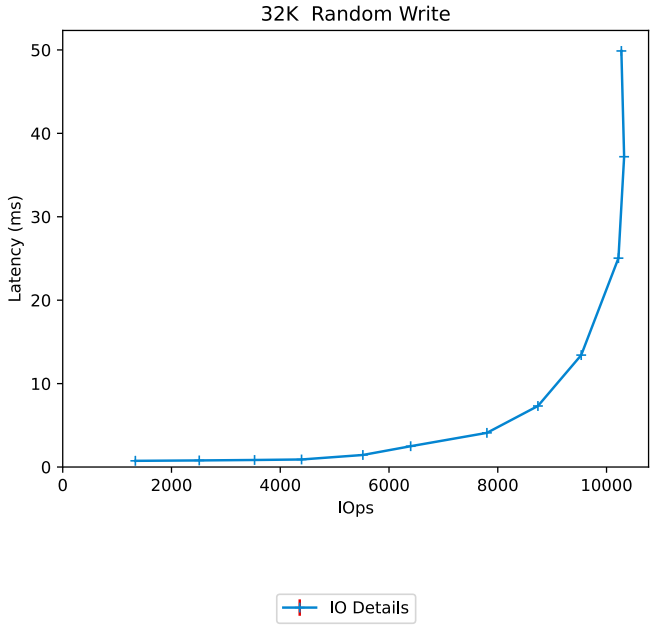


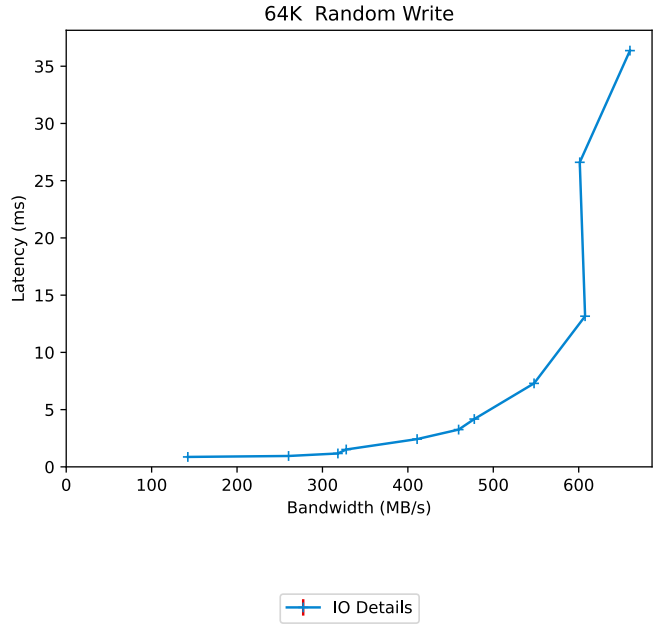
Random Write

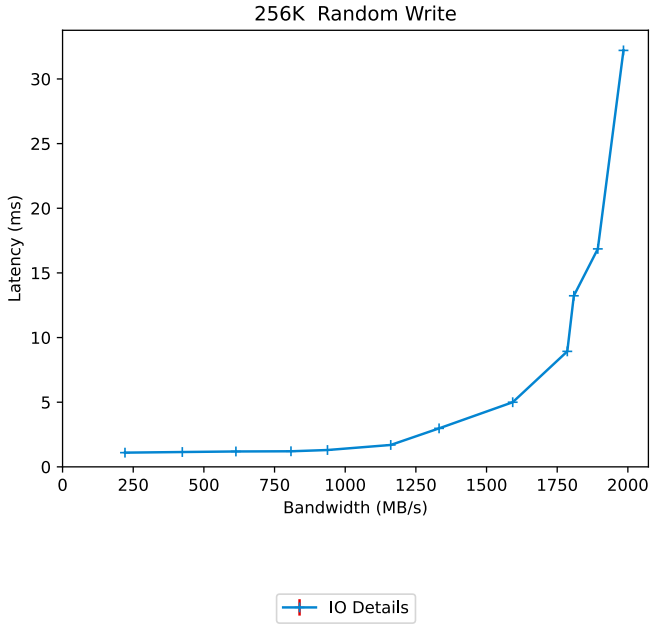

4K Random Write
Latency (ms)
IOps
IO Details

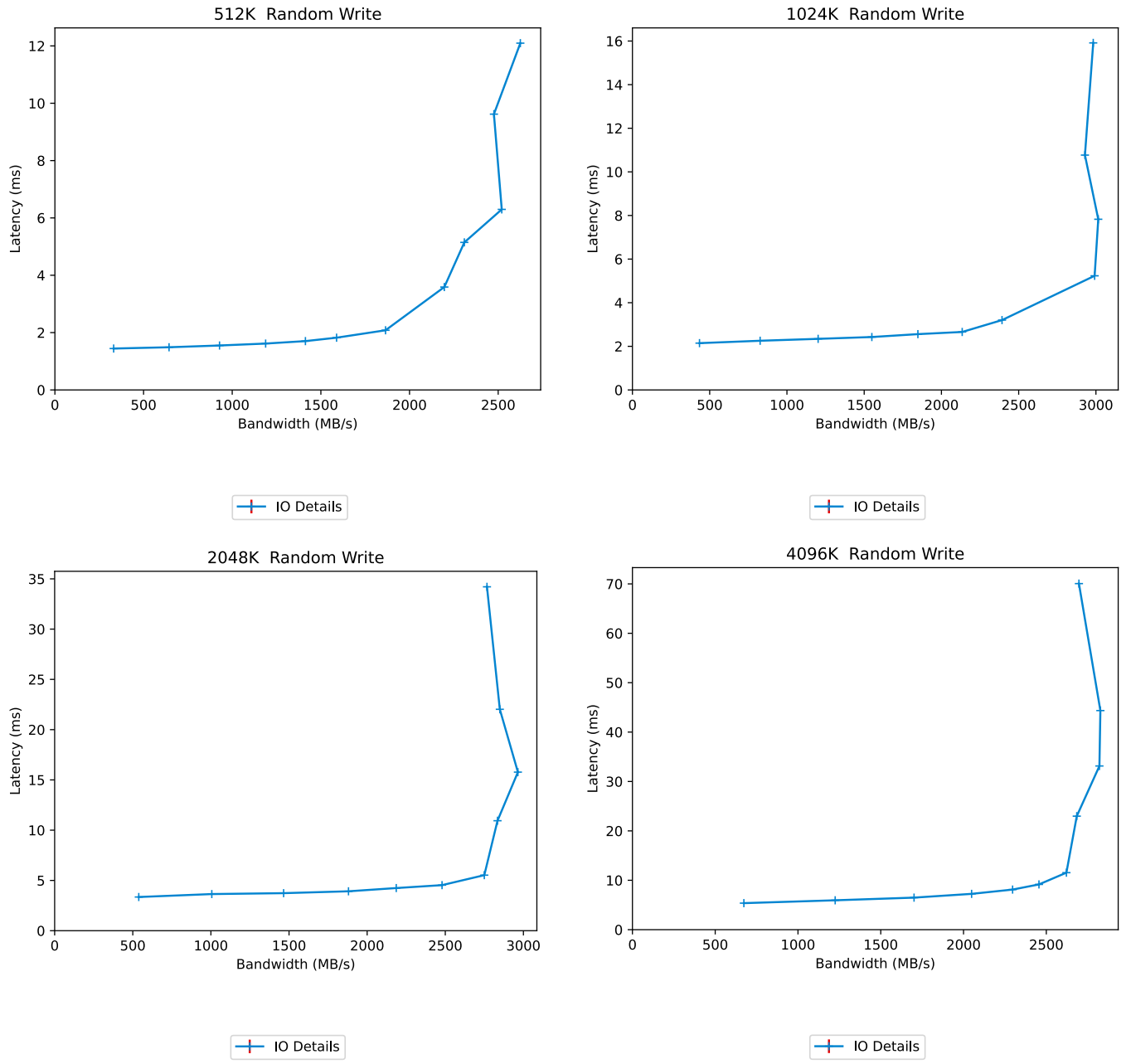

8K Random Write
Latency (ms)
IOps
IO Details


16K Random Write
Latency (ms)
IOps
IO Details

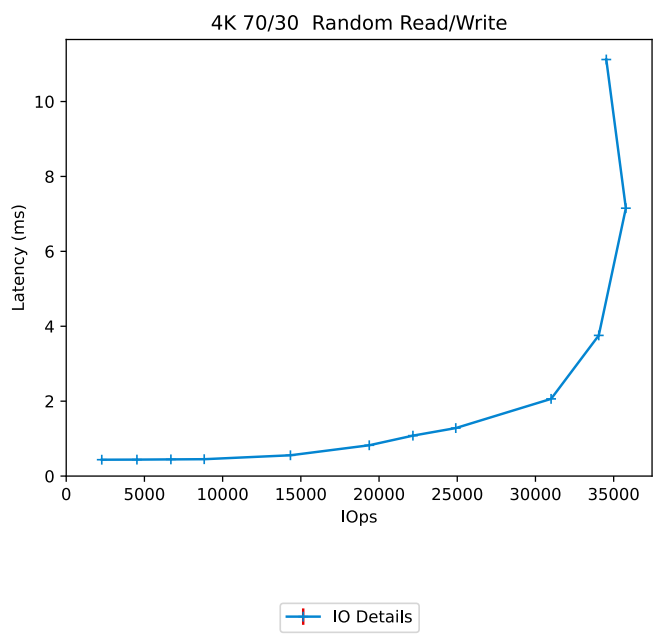

32K Random Write
Latency (ms)
IOps
IO Details

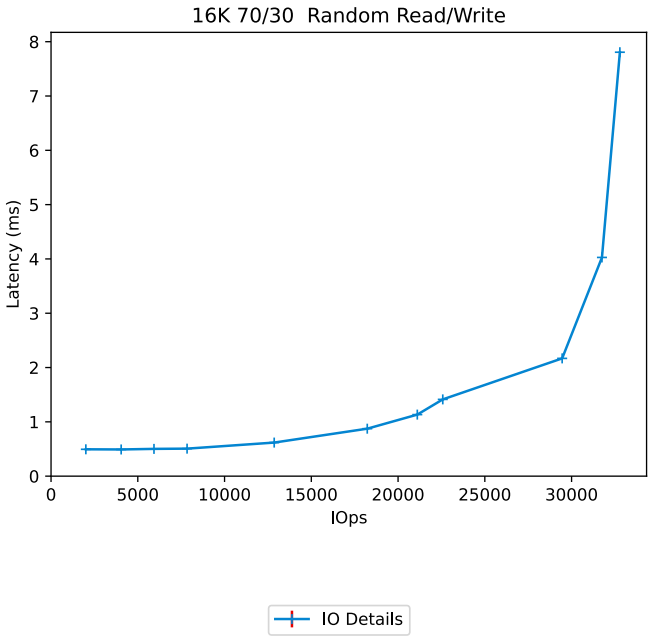

64K Random Write
Latency (ms)
Bandwidth (MB/s)
IO Details

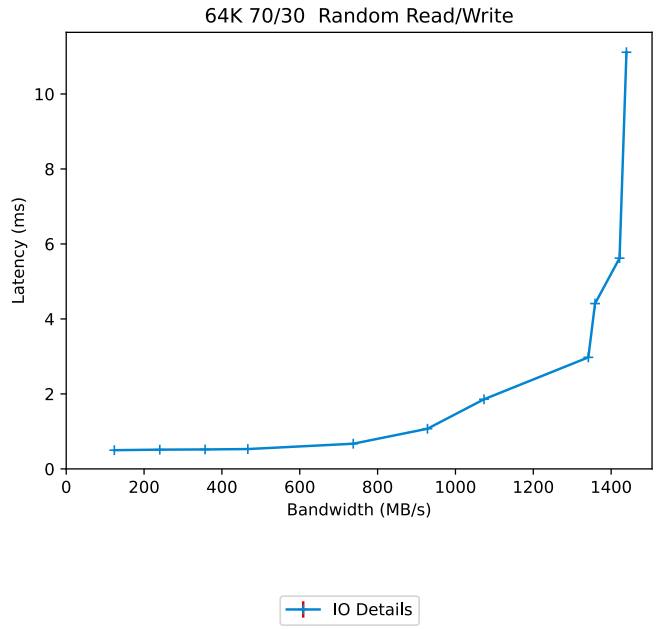

256K Random Write
Latency (ms)
Bandwidth (MB/s)
IO Details

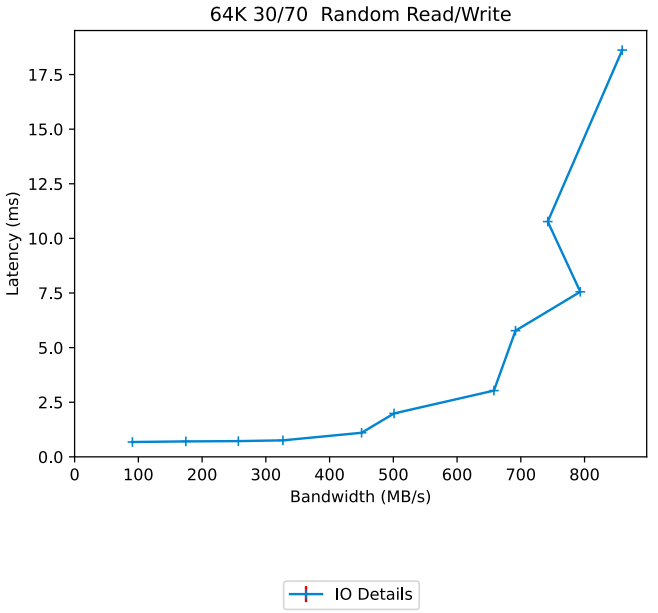


Random Read/Write


4K 70/30 Random Read/Write
Latency (ms)
IOps
0
2
4
6
8
10
0
5000
10000
15000
20000
25000
30000
35000
IO Details


16K 70/30 Random Read/Write
Latency (ms)
IOps
0
1
2
3
4
5
6
7
8
0
5000
10000
15000
20000
25000
30000
IO Details


64K 70/30 Random Read/Write
Latency (ms)
Bandwidth (MB/s)
0
2
4
6
8
10
0
200
400
600
800
1000
1200
1400
IO Details


64K 30/70 Random Read/Write
Latency (ms)
Bandwidth (MB/s)
0.0
2.5
5.0
7.5
10.0
12.5
15.0
17.5
0
100
200
300
400
500
600
700
800
IO Details

Configuration yaml

```
librbdfio:
  cmd_path: /usr/local/bin/fio.lee
  fio_out_format: json
  log_avg_msec: 100
  log_bw: true
  log_iops: true
  log_lat: true
  norandommap: true
  osd_ra:
  - 4096
  poolname: rbd_replicated
  prefill:
    blocksize: 64k
    numjobs: 1
  procs_per_volume:
  - 1
  ramp: 20
  rbdname: cbt-librbdfio
  time: 60
  time_based: true
  use_existing_volumes: true
  vol_size: 92812
  volumes_per_client:
  - 16
  wait_pgautoscaler_timeout: 10
  workloads:
    16k7030:
      jobname: randmix
      mode: randrw
      numjobs:
      - 1
      op_size: 16384
      rwmixread: 70
      total_iodepth:
      - 1
      - 2
      - 3
      - 4
      - 8
      - 16
      - 24
      - 32
      - 64
      - 128
      - 256
    16krandomread:
      jobname: randread
      mode: randread
      numjobs:
      - 1
      op_size: 16384
```

```
  total_iodepth:
  - 4
  - 8
  - 12
  - 16
  - 24
  - 48
  - 64
  - 128
  - 256
  - 384
  - 588
16krandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
  - 1
  op_size: 16384
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 32
  - 64
  - 128
  - 256
  - 384
  - 512
1Mrandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 1048576
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 12
  - 16
  - 20
  - 24
  - 28
  - 32
1Mrandomwrite:
  jobname: randwrite
  mode: randwrite
  numjobs:
  - 1
  op_size: 1048576
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 5
  - 6
  - 8
```

```
    - 16
    - 24
    - 32
    - 48
  1Mseqread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 1048576
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 6
    - 8
    - 10
    - 12
    - 14
    - 16
    - 20
  1Mseqwrite:
    jobname: seqwrite
    mode: write
    numjobs:
    - 1
    op_size: 1048576
    total_iodepth:
    - 2
    - 4
    - 6
    - 8
    - 16
    - 32
    - 48
    - 64
    - 96
    - 128
    - 160
  256krandomread:
    jobname: randread
    mode: randread
    numjobs:
    - 1
    op_size: 262144
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
  256krandomwrite:
    jobname: randwrite
    mode: randwrite
    numjobs:
    - 1
```

```
    op_size: 262144
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 32
    - 64
    - 96
    - 128
    - 256
  2Mrandomread:
    jobname: randread
    mode: randread
    numjobs:
    - 1
    op_size: 2097152
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
  2Mrandomwrite:
    jobname: randwrite
    mode: randwrite
    numjobs:
    - 1
    op_size: 2097152
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 6
    - 8
    - 16
    - 24
    - 32
    - 48
  2Mseqread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 2097152
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 6
    - 8
```

```
    - 10
    - 12
    - 14
    - 16
    - 20
  2Mseqwrite:
    jobname: seqwrite
    mode: write
    numjobs:
    - 1
    op_size: 2097152
    total_iodepth:
    - 1
    - 2
    - 4
    - 5
    - 6
    - 8
    - 16
    - 32
    - 48
    - 64
    - 96
    - 128
  32krandomread:
    jobname: randread
    mode: randread
    numjobs:
    - 1
    op_size: 32768
    total_iodepth:
    - 2
    - 4
    - 6
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
  32krandomwrite:
    jobname: randwrite
    mode: randwrite
    numjobs:
    - 1
    op_size: 32768
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 32
    - 64
    - 128
    - 256
    - 384
    - 512
  4Mrandomread:
    jobname: randread
```

```
    mode: randread
    numjobs:
    - 1
    op_size: 4194304
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 12
    - 16
    - 20
    - 24
    - 28
    - 32
  4Mrandomwrite:
    jobname: randwrite
    mode: randwrite
    numjobs:
    - 1
    op_size: 4194304
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 6
    - 8
    - 16
    - 24
    - 32
    - 48
  4Mseqread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 4194304
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 6
    - 8
    - 10
    - 12
    - 14
    - 16
    - 20
  4Mseqwrite:
    jobname: seqwrite
    mode: write
    numjobs:
    - 1
    op_size: 4194304
    total_iodepth:
    - 1
    - 2
    - 4
    - 5
```

```
    - 6
    - 8
    - 16
    - 32
    - 48
    - 64
    - 96
    - 128
  4k7030:
    jobname: randmix
    mode: randrw
    numjobs:
    - 1
    op_size: 4096
    rwmixread: 70
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 384
  4krandomread:
    jobname: randread
    mode: randread
    numjobs:
    - 1
    op_size: 4096
    total_iodepth:
    - 4
    - 8
    - 12
    - 16
    - 32
    - 48
    - 64
    - 128
    - 256
    - 384
    - 588
    - 768
  4krandomwrite:
    jobname: randwrite
    mode: randwrite
    numjobs:
    - 1
    op_size: 4096
    total_iodepth:
    - 2
    - 4
    - 8
    - 16
    - 32
    - 64
    - 128
    - 256
    - 384
```

```
    - 512
    - 768
  512krandomread:
    jobname: randread
    mode: randread
    numjobs:
    - 1
    op_size: 524288
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 8
    - 16
    - 24
    - 32
    - 40
    - 48
    - 64
  512krandomwrite:
    jobname: randwrite
    mode: randwrite
    numjobs:
    - 1
    op_size: 524288
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 6
    - 8
    - 16
    - 24
    - 32
    - 48
    - 64
  512kseqread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 524288
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 6
    - 8
    - 12
    - 16
    - 20
    - 24
    - 32
  512kseqwrite:
    jobname: seqwrite
    mode: write
    numjobs:
    - 1
    op_size: 524288
```

```
  total_iodepth:
  - 1
  - 2
  - 4
  - 6
  - 8
  - 16
  - 32
  - 48
  - 64
  - 96
  - 128
64k3070:
  jobname: randmix
  mode: randrw
  numjobs:
  - 1
  op_size: 65536
  rwmixread: 30
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 32
  - 64
  - 96
  - 128
  - 256
64k7030:
  jobname: randmix
  mode: randrw
  numjobs:
  - 1
  op_size: 65536
  rwmixread: 70
  total_iodepth:
  - 1
  - 2
  - 3
  - 4
  - 8
  - 16
  - 32
  - 64
  - 96
  - 128
  - 256
64krandomread:
  jobname: randread
  mode: randread
  numjobs:
  - 1
  op_size: 65536
  total_iodepth:
  - 2
  - 4
  - 6
  - 8
  - 12
  - 16
```

```
    - 24
    - 32
    - 64
    - 128
    - 256
  64krandomwrite:
    jobname: randwrite
    mode: randwrite
    numjobs:
    - 1
    op_size: 65536
    total_iodepth:
    - 2
    - 4
    - 6
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 256
    - 384
  64kseqread:
    jobname: read
    mode: read
    numjobs:
    - 1
    op_size: 65536
    total_iodepth:
    - 2
    - 4
    - 6
    - 8
    - 16
    - 24
    - 32
    - 64
    - 128
    - 192
    - 256
  64kseqwrite:
    jobname: write
    mode: write
    numjobs:
    - 1
    op_size: 65536
    total_iodepth:
    - 4
    - 8
    - 12
    - 16
    - 32
    - 48
    - 64
    - 128
    - 256
    - 384
    - 512
  8krandomread:
    jobname: randread
    mode: randread
    numjobs:
```

```
    - 1
    op_size: 8192
    total_iodepth:
    - 4
    - 8
    - 12
    - 16
    - 32
    - 48
    - 64
    - 128
    - 256
    - 384
    - 588
    - 768
  8krandomwrite:
    jobname: randwrite
    mode: randwrite
    numjobs:
    - 1
    op_size: 8192
    total_iodepth:
    - 2
    - 4
    - 8
    - 16
    - 32
    - 64
    - 128
    - 256
    - 384
    - 512
    - 768
  precondition:
    jobname: precond1rw
    mode: randwrite
    monitor: false
    numjobs:
    - 1
    op_size: 65536
    time: 120
    total_iodepth:
    - 16
  seq16kread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 16384
    total_iodepth:
    - 2
    - 4
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 96
    - 128
    - 192
    - 288
  seq16kwrite:
```

```
    jobname: seqwrite
    mode: write
    numjobs:
    - 1
    op_size: 16384
    total_iodepth:
    - 2
    - 4
    - 8
    - 16
    - 32
    - 48
    - 64
    - 128
    - 256
    - 384
    - 512
    - 768
    - 1024
  seq256kread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 262144
    total_iodepth:
    - 1
    - 2
    - 3
    - 4
    - 5
    - 8
    - 16
    - 24
    - 32
    - 48
    - 64
  seq256kwrite:
    jobname: seqwrite
    mode: write
    numjobs:
    - 1
    op_size: 262144
    total_iodepth:
    - 1
    - 2
    - 4
    - 8
    - 16
    - 24
    - 32
    - 64
    - 96
    - 128
    - 256
  seq32kread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 32768
    total_iodepth:
    - 2
```

```
    - 4
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 96
    - 128
    - 192
  seq32kwrite:
    jobname: seqwrite
    mode: write
    numjobs:
    - 1
    op_size: 32768
    total_iodepth:
    - 2
    - 4
    - 8
    - 16
    - 32
    - 64
    - 128
    - 256
    - 512
    - 768
  seq4kread:
    jobname: seqread
    mode: read
    numjobs:
    - 1
    op_size: 4096
    total_iodepth:
    - 2
    - 4
    - 8
    - 12
    - 16
    - 24
    - 32
    - 64
    - 96
    - 128
    - 192
    - 288
    - 384
  seq4kwrite:
    jobname: seqwrite
    mode: write
    numjobs:
    - 1
    op_size: 4096
    total_iodepth:
    - 2
    - 8
    - 16
    - 24
    - 32
    - 48
    - 64
    - 128
    - 256
```

```
        - 384
        - 512
        - 768
        - 1024
        - 1280
        - 1536
      seq8kread:
        jobname: seqread
        mode: read
        numjobs:
        - 1
        op_size: 8192
        total_iodepth:
        - 2
        - 4
        - 8
        - 12
        - 16
        - 24
        - 32
        - 64
        - 96
        - 128
        - 192
        - 288
        - 384
      seq8kwrite:
        jobname: seqwrite
        mode: write
        numjobs:
        - 1
        op_size: 8192
        total_iodepth:
        - 2
        - 8
        - 16
        - 24
        - 32
        - 48
        - 64
        - 128
        - 256
        - 384
        - 512
        - 768
        - 1024
        - 1280
        - 1536
cluster:
  archive_dir: /tmp/cbt
  ceph-mgr_cmd: /usr/bin/ceph-mgr
  ceph-mon_cmd: /usr/bin/ceph-mon
  ceph-osd_cmd: /usr/bin/ceph-osd
  ceph-run_cmd: /usr/bin/ceph-run
  ceph_cmd: /usr/bin/ceph
  clients:
  - --- server1 ---
  clusterid: ceph
  conf_file: /cbt/ceph.conf.4x1x1.fs
  fs: xfs
  head: --- server1 ---
  iterations: 1
  mgrs:
```

```
    --- server1 ---:
      a: null
  mkfs_opts: -f -i size=2048
  mons:
    --- server1 ---:
      a: --- IP Address ---:6789
  mount_opts: -o inode64,noatime,logbsize=256k
  osds:
  - --- server1 ---
  osds_per_node: 8
  pdsh_cmd: /bin/pdsh
  pdsh_ssh_args: -a -x -l%u %h
  pid_dir: /tmp/cbt/pid
  rados_cmd: /usr/bin/rados
  rbd_cmd: /usr/bin/rbd
  tmp_dir: /tmp/cbt
  use_existing: true
  user: root
monitoring_profiles:
  collectl:
    args: -c 18 -oz -sZC -i 10 --sep ";" -F0 -P -f {collectl_dir}
```